

HOW FM WORKS

Carriers & Modulators

RATIO · LEV

- Carriers are heard; modulators reshape a carrier's tone
- Modulator LEV sets brightness; RATIO multiplies the note
- Whole-number ratios sound harmonic, odd ones metallic

Op-Mods (the MOD rows)

MOD1-4

- MODS page: set a slot's DEST to MOD1-4 to feed that op-mod
- An operator's MOD slot applies it to LEV, RAT, PIT or FBK
- One envelope can drive several operators; FM1-FM4 automate

OPERATOR CONTROLS

RATIO	Note multiplier <i>Decimal: 2.00 = octave up, 0.50 = octave down</i>
LEV/FB	Level & feedback <i>Per operator</i>
MOD LEV	Level slot <i>Scales level from 00 up to LEV</i>
MOD RAT	Ratio slot <i>Adds 0.00 to +16.00 to the ratio</i>
MOD PIT	Pitch slot <i>Adds semitones to that operator</i>
MOD FBK	Feedback slot <i>Scales feedback from 00 up to FB</i>
SHAPES	SIN, SW2-SW6, TRI, SAW, SQU, PUL, IMP, NOISE <i>Plus W wavetable shapes</i>
IMP	Impulse <i>FB sets the pulse width</i>
W...	Wavetables <i>FB scans through the table</i>

USEFUL ALGORITHMS

00	A → B → C → D <i>Serial chain: bright, complex stacks</i>
04	A + B + C → D <i>Three modulators into one carrier</i>
08	A → B + C + D <i>One modulator shared by three carriers</i>
0B	A + B + C + D <i>Additive: four carriers, no FM</i>
ALGO	Others <i>The screen draws each routing as you scroll</i>

DRUMS & PERCUSSION

FM Kick

MOD RAT

- A 2-op stack with the modulator at LEV 00: one SIN carrier
- MOD1 → AHD, short DEC → op-mod on the carrier's RAT slot
- MOD2 → AHD on volume; its DEC sets the kick's length

Snare

NOISE · AMP

- A tone carrier plus a NOISE operator for the rattle
- Short envelope on carrier LEV through an op-mod
- Raise AMP (around 20) so clipping makes a stick transient

Bell

ODD RATIO

- Modulator RATIO off the grid (e.g. 3.50), low feedback
- One AHD op-mod on both modulator LEV and carrier LEV
- Brightness fades with the volume, like a struck bell

Hi-Hats

NOISE

- NOISE shape on the carrier, very short AHD on volume
- High-pass FILTER; KIL 02 for closed, longer DEC for open

Toms

MOD RAT

- Start from the FM kick with a higher note and slower RAT decay
- Step through a few notes for a tom fill

Layer Two Sounds

TWO STACKS

- Pick an algorithm with two separate 2-op stacks
- One stack for the body, the other for a click or noise tail

BASSES

Round FM Bass

RATIO 0.50

- Modulator RATIO 0.50 or 0.25 adds sub-like weight
- Short AHD on the modulator's LEV slot for the pluck
- Keep DEL and REV sends low so the low end stays tight

Growl & Wobble

FB · FBK

- Raise the modulator's FB for grit
- LFO → op-mod → the modulator's FBK slot to wobble it
- LIM SIN or POST:AD thickens without harsh clipping

No More Clicks

DE1 · ATK

- Long decays clash with fast notes: shorten them with DE1
- Still clicking? Raise the volume envelope's ATK above 00

CHORDS, PADS & LEADS

Four-Voice Chords

ALGO 0B

- ALGO 0B makes all four operators carriers
- RATIOS 1.00 · 1.26 · 1.50 · 2.00 = major chord plus octave
- Minor: 1.00 · 1.19 · 1.50; balance voices with LEV

Dub Pad

ALGO 08

- One modulator shared by three carriers
- Carriers on a minor chord across octaves (1.00 · 2.38 · 5.99)
- Detune the modulator slightly; slow AHD, MFX and REV sends

PWM Lead

IMP · FBK

- Carrier shape IMP; its FB sets the pulse width
- LFO → op-mod → the carrier's FBK slot sweeps the width

Wavetable Scan

W SHAPES

- Pick a W shape; FB moves through the wavetable
- A slow LFO on the FBK slot evolves the tone over bars

Expressive Brightness

TRACKING

- MOD → TRACKING, SRC VELOCITY → op-mod on modulator LEV
- Soft notes are mellow, hard notes bite, like an electric piano

Detuned Unison Pad

ALGO 0B

- ALGO 0B with RATIOS 1.00 · 1.01 · 0.99 · 2.00
- SAW or SQU shapes, slow ADSR, and an MFX send for chorus

Additive Organ

ALGO 0B

- ALGO 0B with RATIOS 1.00 · 2.00 · 3.00 · 4.00 on SIN shapes
- Each operator's LEV works like a drawbar

Vibrato

LFO · PVB

- MOD → LFO on pitch with a small AMT; FREQ sets the speed
- Or PVB in the phrase for vibrato on single notes

Per-Step Accents

FM1 · DE1

- FM1 on a step changes op-mod 1's amount for that note
- DE1 lengthens or shortens envelope 1's decay per step